ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 2ND YEAR

DATA STRUCTURES AND ALGORITHMS

PAPER 1 · PRACTICE A

MARKS: 100 **TIME: 3 hours**

BSc Computer Science, 2nd year, semester 1 final examination. Section A: objective questions (10) · Section B: algorithm analysis, linked lists, stacks and queues, trees, sorting and hashing, graphs (90).

Written by Academic Central for practice, following second-year Computer Science syllabuses at South African universities (data structures and algorithms). It is not a university examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SIX questions. Answer ALL the questions.
2. Write code in Java. Pseudocode is accepted ONLY where a question says so.
3. Show your working for traces and calculations.
4. Calculators are NOT allowed.
5. Number your answers exactly as the questions are numbered.

SECTION A

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Which data structure works on the last-in, first-out (LIFO) principle?

A Queue
B Stack
C Heap
D Hash table

1.2 What is the worst-case time to search for a key in an UNBALANCED binary search tree with n nodes?

A O(1)
B O(log n)
C O(n)
D O(n log n)

1.3 What is the average time to insert into a hash table with a good hash function and a low load factor?

A O(1)
B O(log n)
C O(n)
D $O(n^2)$

1.4 Which traversal of a binary search tree visits the keys in ascending order?

A Preorder
B Inorder
C Postorder
D Level order

1.5 What is the worst-case time complexity of merge sort?

A O(n)
B O(n log n)
C $O(n^2)$
D O(log n)

1.6 Which data structure does breadth-first search (BFS) use?

A A stack
B A queue
C A priority queue
D A binary search tree

1.7 In a min-heap, the smallest element is stored …

A in a leaf.
B at the root.
C in the last position.
D anywhere.

1.8 Quick sort shows its worst-case behaviour when the pivot is always …

A the median value.
B the smallest or largest value.
C a random value.
D a duplicate value.

1.9 The height of a complete binary tree with n nodes is about …

A n.
B n / 2.
C $\log_2 n$.
D $\sqrt{n}$.

1.10 Which sorting algorithm is NOT stable in its usual form?

A Merge sort
B Insertion sort
C Bubble sort
D Quick sort

[10]

SECTION B

QUESTION 2: ALGORITHM ANALYSIS

2.1 Give the formal definition of f(n) = O(g(n)). (2)

2.2 Give the time complexity, in Big-O notation, of each code fragment. Give a reason for each answer. (3 × 2) (6)

```
// (a)
for (int i = 0; i < n; i++)
    for (int j = 0; j < n; j++)
        sum++;

// (b)
for (int i = 1; i < n; i = i * 2)
    sum++;

// (c)
for (int i = 0; i < n; i++)
    sum++;
for (int j = 0; j < 1000; j++)
    sum++;
```

2.3 Prove that $3n^2 + 5n + 2$ is $O(n^2)$ by finding constants c and n_0. (3)

2.4 Arrange these growth rates from the slowest-growing to the fastest-growing: n log n, 2^n, log n, n^2, n, 1 (2)

2.5 An $O(n^2)$ algorithm takes 2 seconds for n = 1 000. Estimate how long it will take for n = 4 000. Show your working. (2)

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QUESTION 3: LINKED LISTS, STACKS AND QUEUES

A singly linked list of integers uses the class below. The list keeps a reference called head to its first node.

```
class Node {
    int data;
    Node next;
    Node(int data) { this.data = data; }
}
```

3.1 Starting with an empty list, the operations addFirst(5), addFirst(8), addLast(3), removeFirst() and addLast(9) are done in this order. Show the list after the last operation. (3)

3.2 Write an iterative Java method int size() that returns the number of nodes in the list. (4)

3.3 Write a Java method void reverse() that reverses the list in place by changing the next references (do not create new nodes). (5)

3.4 Use a stack to evaluate the postfix expression 6 2 3 + * 4 − Show the contents of the stack after each token is processed. (4)

3.5 A circular queue is stored in an array of size 5 (positions 0 to 4). front is the position of the first item; new items go into the next free position, wrapping around to position 0. The queue starts empty with front = 0. The operations are: enqueue A, enqueue B, enqueue C, dequeue, enqueue D, enqueue E, enqueue F. Give the contents of positions 0 to 4 and the value of front afterwards. (4)

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QUESTION 4: TREES

4.1 Insert the keys 50, 30, 70, 20, 40, 60, 80, 35 (in this order) into an empty binary search tree. Draw the final tree. (4)

4.2 Give the preorder, inorder and postorder traversals of your tree in 4.1. (3 × 2) (6)

4.3 Delete 30 from the tree in 4.1, using the in-order successor. Draw the tree after the deletion. (3)

4.4 What is the height of the tree in 4.1 (count the edges on the longest path from the root to a leaf)? (1)

4.5 What is an AVL tree, and why is it better than an ordinary binary search tree? (2)

4.6 The class TreeNode has the fields int key, TreeNode left and TreeNode right. Write a recursive Java method int countLeaves(TreeNode t) that returns the number of leaves in the tree rooted at t. (4)

[20]

QUESTION 5: SORTING AND HASHING

5.1 Show the array [7, 3, 9, 1, 5] after EACH pass of insertion sort (ascending). (4)

5.2 The array [26, 5, 37, 1, 61, 11, 59, 15] is partitioned once for quick sort, using the LAST element as the pivot (Lomuto partition). Give the array after the partition and the final position (index) of the pivot. (4)

5.3 Explain why merge sort needs extra memory, while quick sort usually does not. (2)

5.4 A hash table has 7 slots (0 to 6), the hash function h(k) = k mod 7 and linear probing. Insert the keys 10, 22, 31, 4, 15, 28, 17 in this order. Show the final table. (5)

5.5 What is the load factor of the table after 5.4? Explain the problem of primary clustering in linear probing. (3)

5.6 Name and briefly describe ONE other method of resolving collisions. (2)

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QUESTION 6: GRAPHS

An undirected graph has the adjacency list: A: B, C · B: A, D, E · C: A, F · D: B · E: B, F · F: C, E

6.1 Give the order in which the vertices are visited by a breadth-first search starting at A. Visit neighbours in alphabetical order. (3)

6.2 Give the order in which the vertices are visited by a recursive depth-first search starting at A. Visit neighbours in alphabetical order. (3)

6.3 For a sparse graph, which uses less memory: an adjacency matrix or an adjacency list? Explain. (2)

6.4 A directed acyclic graph has the edges P→Q, P→R, Q→S, R→S and S→T. Give a valid topological order of its vertices. (2)

6.5 A weighted undirected graph has the edges: AB 4, AC 1, BC 2, BD 5, CD 8, CE 10, DE 2, DF 6, EF 3. Use Kruskal's algorithm to find a minimum spanning tree. List the edges in the order they are added, and give the total weight. (5)

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TOTAL: 100

END OF QUESTION PAPER